JavaScript is a trademark originally owned by Sun Microsystems and later acquired by Oracle. It is defined and standardized by ECMA International, a global standards organization–an open standard. The specification that defines JavaScript is called ECMAScript (ES), serving as the language's foundation. The evolution of ECMAScript is managed by TC39, a committee of experts under ECMA International. TC39 follows a rigorous multi-stage process to propose, review, and finalize new features to ensure consistency and compatibility across implementations.

The TC39 process manages the evolution of ECMAScript, with proposals progressing through stages:

- *Stage 0 (Strawperson)*—Initial ideas or input for the specification.
- *Stage 1 (Proposal)*—High-level API proposals are reviewed.
- *Stage 2 (Draft)*—Syntax and semantics are precisely described using formal specification language.
- *Stage 3 (Candidate)*—Designated reviewers approve the current specification text.
- *Stage 4 (Finished)*—The feature is ready for inclusion in the ECMAScript standard.

Once a proposal reaches Stage 4, it is included in the next revision of ECMAScript during its annual ratification process. The latest ECMAScript standard can be accessed at https://tc39.es/ecma262/.

The TC39 process closely mirrors the product development lifecycle, ensuring that new JavaScript features undergo a structured and rigorous evaluation before becoming part of the standard. Figure A.1 illustrates these parallels, showing how both processes move from initial ideas and high-level concepts through detailed design, testing, and refinement, ultimately leading to a final, production-ready release. This alignment guarantees that new language features are thoroughly vetted, tested, and refined, promoting reliability and stability before widespread adoption.

Figure A.1 JavaScript finished proposals